Chapter 2 — Description of Motion

2.1 Functions from $\mathbb{R}$ to $\mathbb{R}^n$

$$\begin{aligned}
\mathbf{x}(t) &= (x_1(t), x_2(t), \ldots, x_n(t)) \\
\mathbf{x}'(t) &= \frac{d\mathbf{x}}{dt} = (x_1'(t), x_2'(t), \ldots, x_n'(t)) \\
\|\mathbf{x}'(t)\| &= \sqrt{(x_1'(t))^2 + (x_2'(t))^2 + \cdots + (x_n'(t))^2} \\
s(t) &= \int_{t_0}^{t} \|\mathbf{x}'(u)\|\, du
\end{aligned}$$

2.1.5 Arc Length Parameterization

$$\begin{aligned}
\frac{d\mathbf{x}}{ds} &= \mathbf{T}(s) \\
\frac{d\mathbf{T}}{ds} &= \kappa(s)\mathbf{N}(s)
\end{aligned}$$

2.1.6 Frenet–Serret Formulas

$$\begin{aligned}
\mathbf{T}'(s) &= \kappa(s)\mathbf{N}(s) \\
\mathbf{N}'(s) &= -\kappa(s)\mathbf{T}(s) + \tau(s)\mathbf{B}(s) \\
\mathbf{B}'(s) &= -\tau(s)\mathbf{N}(s)
\end{aligned}$$

2.1.7 Curvature and Torsion

$$\begin{aligned}
\kappa &= \frac{\|\mathbf{v} \times \mathbf{a}\|}{\|\mathbf{v}\|^3} \\
\tau &= \frac{\mathbf{a}' \cdot (\mathbf{v} \times \mathbf{a})}{\|\mathbf{v}\|^6 \kappa^2} = -\frac{\mathbf{x}''' \cdot (\mathbf{x}'' \times \mathbf{x}')}{\|\mathbf{x}'\|^6 \kappa^2}
\end{aligned}$$

The Darboux vector $\boldsymbol{\omega}$ describes the instantaneous angular velocity of the moving Frenet frame $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ along a space curve.

$$\frac{d\mathbf{T}}{ds} = \boldsymbol{\omega} \times \mathbf{T}, \quad \frac{d\mathbf{N}}{ds} = \boldsymbol{\omega} \times \mathbf{N}, \quad \frac{d\mathbf{B}}{ds} = \boldsymbol{\omega} \times \mathbf{B}$$

where $\boldsymbol{\omega}$ is defined by

$$\boldsymbol{\omega} = \tau\mathbf{T} + \kappa\mathbf{B}$$

Hence, substituting into the Frenet–Serret formulas:

$$\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N}, \tag{1}$$

$$\frac{d\mathbf{N}}{ds} = -\kappa\mathbf{T} + \tau\mathbf{B}, \tag{2}$$

$$\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}. \tag{3}$$

2.1.8 Helix Parameterization

If curvature κ and torsion τ are constants, then

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{\kappa}{\kappa^2+\tau^2}\left[(1-\cos(\sqrt{\kappa^2+\tau^2}\,t))\,\mathbf{u}_1 - \sin(\sqrt{\kappa^2+\tau^2}\,t)\,\mathbf{u}_2\right] + t\,\frac{\tau}{\sqrt{\kappa^2+\tau^2}}\,\mathbf{u}_3$$

Constants for radius and pitch:

$$r = \frac{\kappa}{\kappa^2+\tau^2}, \quad c = \sqrt{\kappa^2+\tau^2}, \quad b = \frac{\kappa}{\kappa^2+\tau^2}$$

$$\left|\frac{1-n}{n}+1\right| = \left|\frac{1-n}{n}+\frac{n}{n}\right| = \left|\frac{1-n+n}{n}\right| = \left|\frac{1}{n}\right|$$

Since n is a positive integer, $|1/n|$ is just $1/n$. **This is the key** We found that the distance from our sequence to the limit is just **$1/n$**. Let's check this: * At $n = 10$, the term is $\frac{1-10}{10} = -0.9$. The distance to -1 is $|-0.9-(-1)| = 0.1$. And $1/n$ is $1/10 = 0.1$. It works * At $n = 100$, the term is $\frac{1-100}{100} = -0.99$. The distance to -1 is $|-0.99-(-1)| = 0.01$. And $1/n$ is $1/100 = 0.01$. It works again2ẆIn the Game (Finding N) Now we can respond to the skeptic's challenge. * **Skeptic's Challenge:** "Make your distance less than my tolerance ϵ" * **Our Task:** We need to find n large enough so that: